

Kundai Farai Sachikonye

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Computational scientist and software engineer: languages and compilers, GPU computing, AI systems, and scientific computing — ideas turned into deployed tools.

Profile

Computational scientist and engineer spanning programming-language and compiler design, GPU / graphics, AI / ML systems, and scientific computing. I consistently take work from concept to deployed, documented, open-source tool.

Technical Skills

Software engineering	Rust, TypeScript, Python; programming-language and compiler design (lexers, parsers, codegen); full-stack (Next.js); async / REST; clean modular architecture
GPU & graphics	WebGPU, WebGL2 / GLSL shaders; GPU compute; computer vision / image analysis
AI / ML	PyTorch; agentic / multi-agent AI; foundation-model adaptation (LoRA); RAG and reasoning pipelines
Data & HPC	Ray, Dask, SLURM; Docker, Kubernetes; large-scale (>100 GB) pipelines
Developer tooling	VSCode extensions including a language server (LSP)

Experience

2021–present	Independent Computational Researcher & Software Engineer, Bitspark GmbH Designed and built compilers / domain-specific languages, GPU pipelines, and AI systems; all deployed and open-source. Rust, Python / PyTorch, TypeScript, distributed / GPU.
2019–2021	Computational Science — Bioinformatics, HPC & Teaching, Technische Universität München, Munich Large-scale data analysis and ML pipelines at HPC scale (PyTorch, SLURM). Taught at Master's level and supervised student work.
2017–2018	Data Pipelines, Universitätsklinikum Hamburg-Eppendorf, Hamburg Automated, reproducible high-throughput data-processing pipelines.

Selected Projects

kwasa-kwasa	A polyglot programming language with its own compiler and a suite of 5 VSCode extensions (incl. a language server). Rust + TypeScript. github.com/fullscreen-triangle/kwasa-kwasa
purpose	Rust CLI (Cargo workspace): an agent-queryable context tool that cuts LLM token burn; trait-based extensible architecture. github.com/fullscreen-triangle/purpose
hieronymus	GPU-accelerated microscopy image analysis (computer vision); WebGL / GLSL. hieronymus-omega.vercel.app

systems- Domain-specific language compiling to GLSL + JS; GPU shader pipeline. systems-biology-
biology- shaders.vercel.app
shaders

Education

2018–2019 **MSc Computational Biology**, *Universität Tübingen*
2014–2017 **MSc Pharmaceutical Biotechnology**, *HAW Hamburg*
2011–2014 **BSc Biochemistry and Cell Biology**, *Jacobs University Bremen*
Additional: doctoral research, computational lipidomics, TUM (not completed).

Languages

English Native / Fluent (C2)

German Conversational

resident in Germany since 2011